

Strange Function

Problem Description

Here is a function:

$$F(x) = 6 * x^7 + 8 * x^6 + 7 * x^3 + 5 * x^2 - y * x \quad (0 \leq x \leq 100)$$

Given a real number y , can you find the function's minimum value when x is between 0 and 100?

Input

The first line of the input contains an integer T ($1 \leq T \leq 100$) which means the number of test cases.

Then T lines follow, and each line has one real number Y ($0 < Y < 1e10$).

Output

The minimum value (accurate up to 4 decimal places), when x is between 0 and 100.

Sample Input

```
2
100
200
```

Sample Output

```
-74.4291
-178.8534
```